

Marco & Polo Go Round

Immersive Virtual Reality

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Item 7

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ABSTRACT

Marco Polo Go Round is a comedic love story with a very surreal twist in which the user is invited to participate in a couple's relationship as their world literally falls apart around them.

It is a narrative experience, set in one location that lasts 14 minutes. Motion-captured actors will drive detailed animated characters such that intimate and touching performances exist within a fully immersive 6DOF world.

KEYWORDS

Immersive, narrative, gravity, unreal, game engine, photogrammetry, scanning, virtual reality

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1 INTRODUCTION

The user is an objective outside participant in our story. This voyeuristic approach, which can be discomforting at first, allows direct access to the emotions our characters are experiencing.

Imagine being invited to stand on stage with actors during a live theater performance only set elements could move with the impossible visual wonder of cinema. In Marco & Polo Go Round the user has agency to move within the room and around the actors and can touch the objects that swirl and crash around them. Because of this proximity to character and the impossible, physical experience of the world turning upside down Marco Polo Go Round becomes a combination of the magic of cinema with the amazing intimacy of live theater. The tactile and the impossible come together to create a new form of entertainment.

The character arcs in MPGR are emotionally complex but the physical experience of entering a dreamscape in which gravity gets turned on its head is mesmerizing. This combination of character driven storytelling with a visceral, 'impossible' experience has not been seen in VR before and legitimizes VR as a space for mature, compelling dramatic experiences.

The user's gaze is guided through the use of character blocking and sound design, as well as carefully planned camera movements. The experience is fully stereoscopic, 6DOF, room scale experience. The target age group is young adults and older adults who seek out character driven and viscerally compelling narrative experiences.



The story takes place in a kitchen. The user moves around Marco and Polo, our main characters, such that proximity to the characters remains dynamic and emotional. The user will even float up with them as they reach the ceiling, physically experiencing the sensation of becoming weightless. Throughout the experience the user can stand and walk around the action.

2 DESIGN AND IMPLEMENTATION

The world of Marco Polo is immersive. The user has agency to move with and around the characters and the gorgeous visual chaos of seeing the world turned upside down. For this reason the experience exists in a CGI environment.

Our experience was designed and built in Unreal. Models of actors were made using photogrammetry. Actors' performances were motion captured. Characters were rigged and animated in Maya. Sims were done in Houdini. The final experience is rendered in real time.

Marco Polo Go Round will push technological boundaries in creating digital VR characters that move with the complexity and emotional depth of real actors. Immersion and 6DOF is what makes Virtual Reality an experience rather than a way of watching a film in 360 degrees. With Marco Polo Go Round, technology is used to create engaging, moving performances designed specifically for an adult audience in ways we have never seen before.

3 CONCLUSION

What makes Marco Polo different than other VR experiences is its aim to present mature, emotional subject matter within the context of a visceral, astounding physical experience. Marco Polo Go Round is a touching, surreal love story and a new way of seeing VR as a space for creating important, moving stories that also take audiences on a mesmerizing journey.